

PlanetScope Reference Manual

A comprehensive guide to PlanetScope satellite data, products, and access methods.

Table of Contents

1. GENERAL INFORMATION	2
2. USEFUL LINKS	2
3. PLANETSCOPE PRODUCT INFORMATION	3
3.1. Satellite Constellation	3
3.2. Instruments	3
3.3. Products	4
3.3.1. PlanetScope basemap mosaic tiles	4
3.3.2. PSScene	4
Asset Types	4
Product Bundles (Order API)	5
4. PROCESSING	6
4.1. Sensor and Radiometric Calibration	6
4.2. Surface reflectance processing	7
4.3. Cloud detection	7
4.4. Harmonization and Normalization	8
4.5. Compositing	8
4.6. Publishing States	9
5. ORDER & DOWNLOAD	10
5.1. Tools	10
5.2. Download using API	10
5.3. Explorer and feature manager	12
5.4. QGIS Plugin	15
6. Questions	17

1. GENERAL INFORMATION

Data catalog

- PlanetScope
- SkySat
- Tanager
- RapidEye
- Analysis-Ready PlanetScope
- Basemaps
- Usable Data Mask

License Model

- Area Under Management License: based on surface area under agreement
- Example: ordering 50 km² grants access to all available data within that area
- Re-downloading previously ordered data does not consume additional quota
- Paying for different data products, in our case we have access to PlanetScope data

General Information

- Almost daily coverage worldwide
- 3.7-4.1 m pixel size (resampled to 3 m)
- Blue, Green, Red, Near Infrared
- additional bands Coastal Blue, Green I, Yellow, and Red Edge
- constellation of > 100 satellites called Doves
- Each Dove satellite is a CubeSat made of three cubic units
- Can be downloaded via Planet explorer ([Planet Explorer](#)), Order API ([Orders Overview | Planet Documentation](#)), Data API ([Data API Overview | Planet Documentation](#))

2. USEFUL LINKS

[PlanetScope | Planet Documentation](#)

[Asset Share - Planet Asset Library \(PAL\)](#) – Technical Information

[notebooks/jupyter-notebooks/workflows/udm2/udm.ipynb at master · planetlabs/notebooks · GitHub](#)

[UDM2 Mask Classification, PlanetScope | Sentinel Hub custom scripts](#)

[APIs | Planet Documentation](#)

[Introduction to Planet's Data and Orders APIs](#) - Tutorial of how to download Planet data via API in Jupyter Notebook

[Flexible Pricing for Satellite Imagery & Data | Planet](#) – different tiers and pricing models

3. PLANETSCOPE PRODUCT INFORMATION

- Two geometry types: basic (not orthorectified) and ortho scenes (additional post processing applied)
- Two imagery options: visual (RGB) with color correction for visual interpretation and analytic products (4 or 8 band) as TOA radiance or SR

3.1. Satellite Constellation

- Multiple launches of groups of satellites
- PlanetScope is a constellation of ~ 118 satellites
- Publication of image depends on the quality of the images and the ground data for registration
 - Less accurate over open water, mountainous terrain or cloudy areas

3.2. Instruments

Instrument	ID	Description	Interoperable with Sentinel-2
Dove Classic	PS2	4-band imagery (Jul 2014 – Apr 2022)	Yes, all 4 bands
Dove-R	PS2.SD	4-band imagery (Mar 2019 – Apr 2022)	
SuperDove	PSB.SD	8-band imagery (since 2020)	6 from 8 bands (not green I and yellow)

3.3. Products

3.3.1. PlanetScope basemap mosaic tiles

[PlanetScope Basemaps Technical Specification, July, 2024](#)

- Resolution of 4.77m or 2.39m → improves at higher and lower latitudes
- Can be viewed within the Planet graphical user interface
- Visual or Surface Reflectance
- WGS84 Web Mercator (EPSG: 3857)
- Generated weekly, monthly, quarterly intervals

Visual basemaps

- Minimize cloud cover, haze and topographic variation effects
- Color-corrected
- RGB

Surface reflectance basemaps

- SR imagery asset type (analytic_sr)
- Reduce variability caused by atmospheric effects
- To perform spectral, quantitative analysis as well as time series

3.3.2. PSScene

[PSScene | Planet Documentation](#)

Individual framed scenes; 3-, 4-, and 8-band imagery; basic & orthorectified (corrected for terrain distortions, projected to 4326)

Asset Types

Asset Type	Description	Usage	Asset
Basic Analytic	Unorthorectified, calibrated imagery corrected for sensor artifacts; TOA radiance 16-bit scaled	users who wish to geometrically correct the data themselves	<i>basic_analytic_4b,</i> <i>basic_analytic_8b,</i>

TOA Radiance	Radiometrically-calibrated, multispectral imagery; corrected for sensor & terrain distortions 16-bit scaled radiance	analytic applications which require imagery with accurate geolocation and cartographic projection	<i>ortho_analytic_4b, ortho_analytic_8b</i>
Visual	Orthorectified, RGB imagery optimized for visualization	for simple and direct visual inspection; used for Geographic Information System or applications	<i>ortho_visual</i>
Surface Reflectance	Orthorectified, radiometrically corrected imagery	for temporal analysis and monitoring applications, especially in agriculture and forestry sectors	<i>ortho_analytic_4b_sr, ortho_analytic_8b_sr</i>
Usable Data mask	Unorthorectified usable data mask	Cloud, Shadow, haze.. masking	<i>Basic_udm2</i>

Naming convention:

- *{acquisition-date}_{acquisition-time}_{acquisition-time-seconds-hundredths}_{satellite-id}_{product-level}_{band-product}.{ext}*

Example:

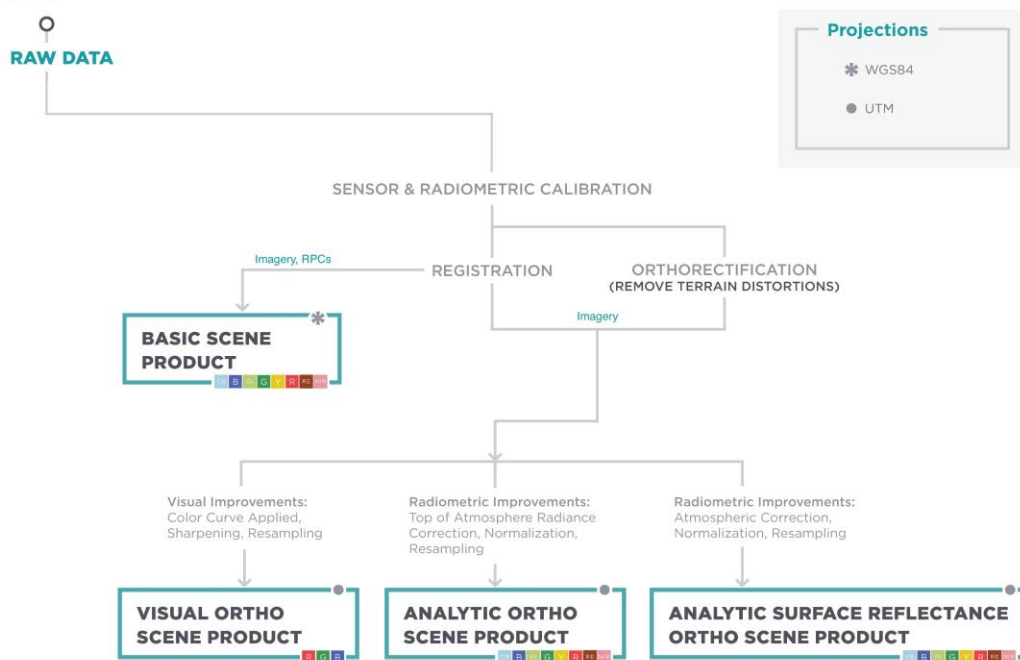
- 20230207_143613_03_241c_3B_AnalyticMS_SR_8b.tif

Product Bundles (Order API)

Bundle	Contents
analytic	4-band (Blue, Green, Red, NIR), 16-bit
analytic_sr	Surface reflectance version of analytic
analytic_xml	Metadata in XML
analytic_udm2	Unusable Data Mask (UDM2)
visual	3-band RGB, 8-bit, contrast-stretched
basic_analytic	Unrectified imagery

4. PROCESSING

IMAGE PROCESSING CHAIN



4.1. Sensor and Radiometric Calibration

Correction	How
Darkfield / Offset Correction	Correction for sensor bias and dark noise
Flat field correction	- Fixes non-uniform brightness across the image - Collection of fields for every optical instrument - Used to correct image lighting and CCD effects
Camera acquisition parameter correction	- Determination of common radiometric response for each image - Standardization of image brightness resulting from different camera settings

Absolute calibration	Transformation of DN's to physical-based radiance values
Orthorectification	- Removes terrain distortions - Terrain model derived from multiple sources (Intermap, NED, SRTM, other local elevation datasets)

4.2. Surface reflectance processing

- Top of Atmosphere (TOA) reflectance calculation using coefficients supplied with the at-sensor radiance product.
- Lookup table (LUT) generation using radiative transfer code and MODIS near-real-time data inputs.
- Conversion of TOA reflectance to surface reflectance for all combinations of selected ranges of physical conditions and for each satellite sensor type using its individual spectral response and estimates of the state of the atmosphere.

4.3. Cloud detection

➔ UDM2.0 for all images before November 29,2023

➔ UDM2.1 afterwards

- Based on thresholding on band reflectance's and indices values
- UDM2 is based on supervised machine learning that use observation data from Planet satellites to train a classification model
- modified UNET: deep learning supervised segmentation model, running on 4-band (RGB-NIR) top-of-atmosphere radiance imagery
- hand-labeled PLANETSCOPE and Skysat images to mark each pixel into one of the 5 classes
- Classes:
 - Clear
 - Snow
 - Shadow
 - Haze light
 - Haze heavy
 - Cloud
 - Confidence
 - UDM1

4.4. Harmonization and Normalization

Scene Level Normalization of Planet Dove Imagery

- ➔ Option to harmonize PlanetScope bands to S2 bands using the harmonization tool in the Orders API
- ➔ In addition, radiometrically correct imagery captured by different generations of PlanetScope instruments, i.e. PS2, PS2.SD, PSB.SD
- Correction of different relative spectral responses caused by multiple PlanetScope constellations
- corrects for inter-scene variability within the same generation of sensor
- Based on spectral response relationships and / or multi-temporal reference sampling
- regularization metric to minimize changes in band ratios during harmonization
- based on relationship between Planet and Sentinel band, not matching acquisitions

Normalization

- scene level normalization method minimizes scene-to-scene and sensor-to-sensor variability by matching each scene to a Sentinel2-based seasonal reference dataset
 - fit a linear model for each band in each scene based on co-located, non-cloudy PlanetScope and seasonal model reference pixels
- minimize large changes in band ratios during normalization
- transform input Planetscope reflectance coefficients to approximately match Sentinel2 spectral response and minimize large deviations between scenes from different days and ambient conditions

4.5. Compositing

- Merges multiple scenes into one by overlaying input images on top of each other and flattens them together, with later images in the list applied on top of earlier ones
- Optimizes quota usage by eliminating unnecessary overlap between scenes, it reduces file size and avoids redundant data
- Corresponding Usable Data Masks (UDMs) are composited alongside the imagery, with _composite appended to output filenames

- Delivers metadata file is included with cloud statistics (e.g., cloud_cover) and an updated geometry reflecting the composite extent
- order (default), Produces a single composite from all items in the order
- strip_id, Groups items by strip and produces one composite per strip

4.6. Publishing States

Preview

- images are unrectified; metadata changes are expected upon subsequent republishing

Images that are not passed through all processing steps:

- Gaps in the collections before 19th July 2023 caused by cloud cover, weather conditions
- After that, likely caused by band misalignment, missing pixels, absence of image collection over the area
- Challenging topography, open water lead to problems in georeferencing

Standard

- Data is stable and rectified, but subject to minimal metadata changes upon republishing.

Finalized

- Data is stable and rectified, no additional metadata changes are expected

5. ORDER & DOWNLOAD

- Subscription: 12 months, starting at first purchase
- Land under management license
- AOIs must be in EPSG:4326
- Methods: Planet Explorer, QGIS Plugin, API

5.1. Tools

Explorer

- ➔ Browser-based interface for searching and ordering data
- Upload AOIs (EPSG:4326)
- Apply filters (date, cloud cover)
- Preview and order scenes

QGIS Plugin

- Define AOI, filter by date/cloud cover
- Search and order products (only 4-band SR GeoTIFFs)
- Supports clipping and composites

API

- Order multiple scenes at once
- Clip area to AOI
- Apply filters
- Apply harmonization

5.2. Download using API

- You can use quick search for just searching for images in a specific time and with specific properties like cloud cover
- The image ids of that search are then used for ordering the data you need
- You can download data via order or data API → both differ (see below)
- Bulk download possible up to 500 images
- Continues download possible

Don't use the data api because you can't clip to your AOI!!!

Method	What happens	Use Case	Credits Used?	Clipping?
Quick Search	Lists full scenes matching your filters		✗	✗
Data API	Prepares the full scene or analytic asset for download	Best for browsing scenes, fetching metadata, downloading individual assets	✓	✗
Order API	Apply operations like clipping to AOI	Best for bulk downloads, automating workflows, and custom image processing	✓ only for AOI area	✓
Subscription API	Create subscription to regularly download data for an AOI	Continuous access, monitoring, updates	✓	✓

- ➔ R Script to order and download data
Environwin/FORCE/scripts/RScripts/automated_download.R
- ➔ Downloaded data is saved under Environwin/PLANET and afterwards stored in CEPH_BASEDATA
- ➔ Data is ordered and downloaded in “batches”, otherwise if trying to order a lot of data at the same time, the process fails
- ➔ In the Script, it is possible to select the testsite from a shapefile and extract the coordinates to transform to JSON format which is needed for the API
- ➔ Download data for a longer period, e.g. one full year or just for a subset of the year – has to be a continuous period, it is not possible to download only discontinuous month, e.g. only February, October and November.

Product bundles for Order API

Bundle Name	Contents
<i>analytic</i>	4-band imagery (Blue, Green, Red, NIR) in 16-bit

<i>analytic_sr</i>	Surface reflectance version of analytic
<i>analytic_xml</i>	Includes metadata XML with analytic
<i>analytic_udm2</i>	Includes Unusable Data Mask (UDM2) with analytic
<i>visual</i>	3-band RGB (8-bit, contrast-stretched)
<i>basic_analytic</i>	Unrectified imagery (for scientific uses or calibration)
<i>basic_analytic_dn</i>	Digital Number version of unprocessed 4-band data

PlanetScope Asset Types

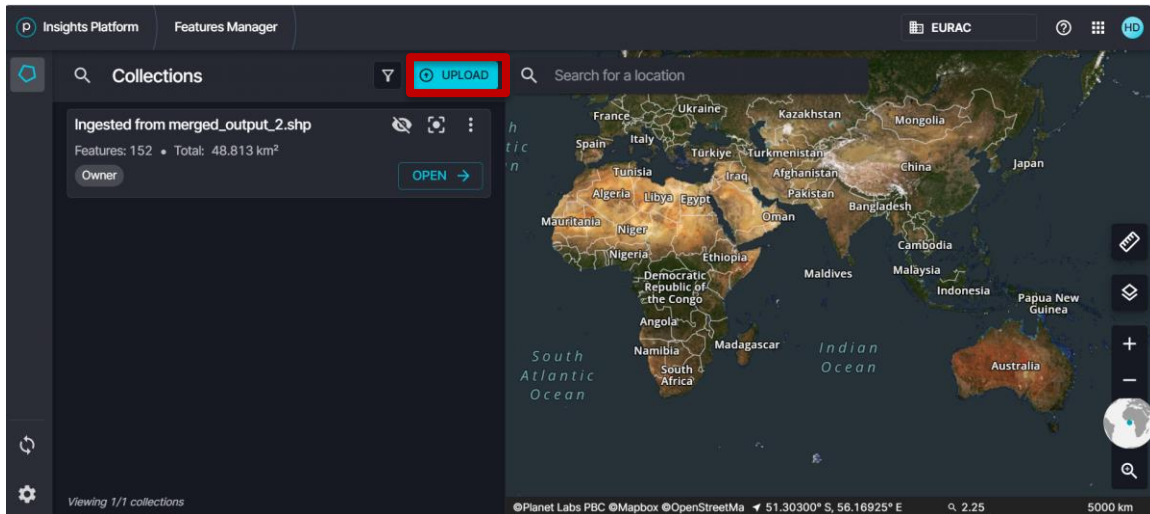
Asset Type	Description
<i>analytic</i>	TOA - 4-band image (16-bit)
<i>analytic_sr</i>	Surface reflectance-corrected 4-band image
<i>visual</i>	RGB-only image (8-bit)
<i>udm</i>	Usable data mask
<i>udm2</i>	More detailed mask (e.g., cloud, haze, shadow info)
<i>metadata</i>	Scene-level metadata in JSON
<i>analytic_xml</i>	XML-formatted metadata
<i>thumbnail</i>	Small preview image (RGB)

5.3. Explorer and feature manager

[Features Manager](#)

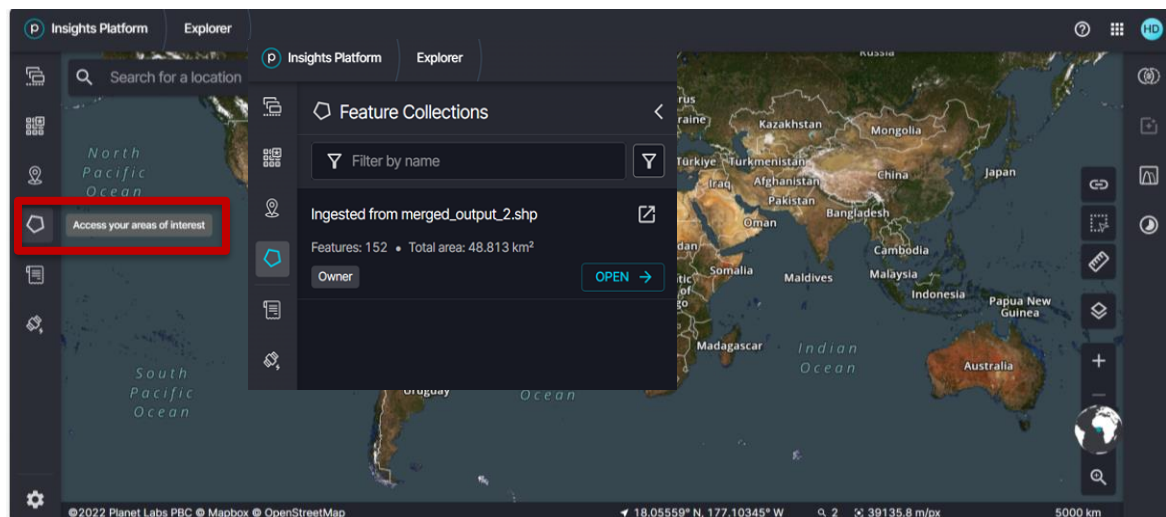
[Browser • Planet Insights Platform](#)

- Feature manager allows upload of multiple areas of interest
- Allows to use the AOIS as well in the Browser to search for images
- Must be in WGS 84, EPSG: 4326



Feature manager

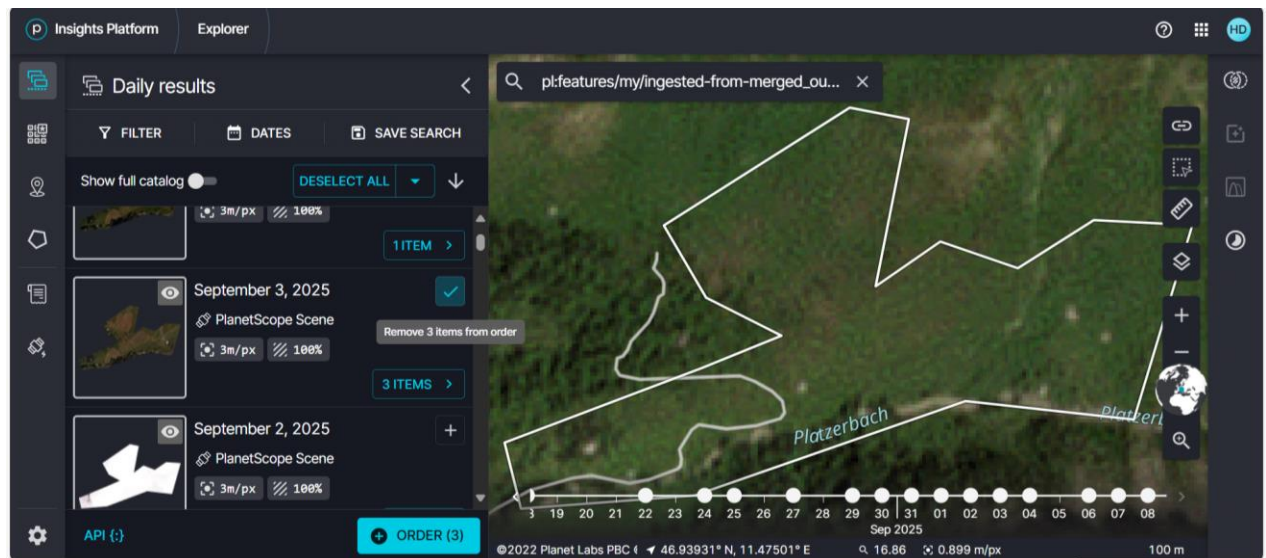
- Upload aois to feature manager -> has to be in WGS 84, EPSG: 4326
- Open features and select polygon to search for images
- Click on the three dots and click "Search here"
- This will open the planet explorer and will search for images already cropped to the aoi
- Apply filter e.g. date and cloud cover
- Possibility of previewing images



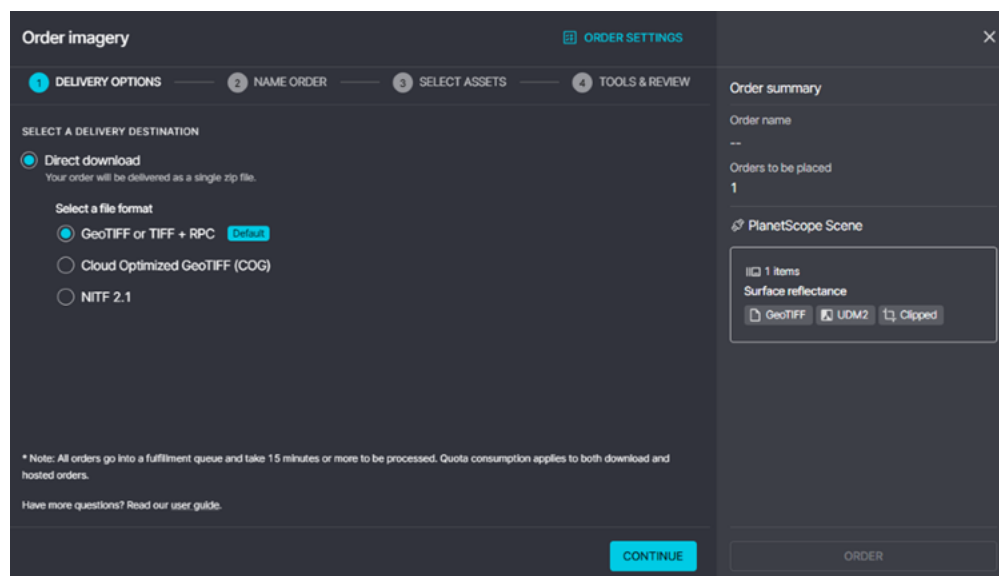
Explorer

Order scenes

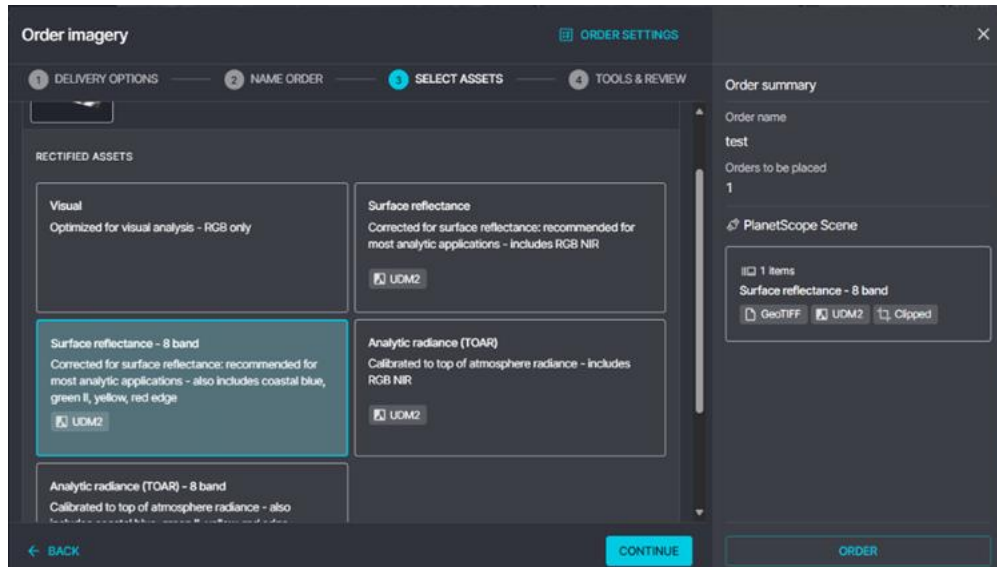
- Select image(s) and order scenes
- Select download format
- Give order a name
- Select the asset --> 8 band surface reflectance
- In the last step select Clip Items to AOI



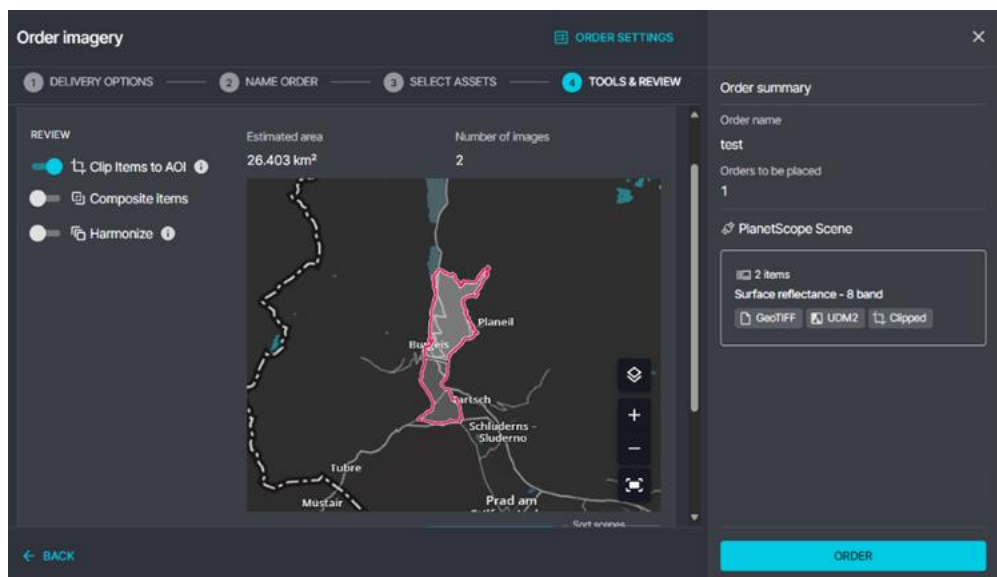
Explorer → Add scenes to order list



Select delivery options, then name your order



Select the product



Apply a clip to your image

5.4. QGIS Plugin

[Planet QGIS Plugin | Planet Documentation](#)

- Define area of interest, e.g. draw rectangle
- Filter scenes, e.g. date range and cloud cover
- Search results
- No preview available

- Always name the order to find it in the "My order" section
- Select the asset (product) that you want to download
- In the last step define if you want to clip the image to the aoi, if you want to have a composite of the images, ...

1 – Search for images

Planet Explorer [2.3.4]

Preview, Download or Stream Planet Imagery

Daily Imagery Basemaps

Select an area of interest or [load a saved search](#)

▼ Area of interest

Extent Draw Selection Upload

524186,46.75139],[10.521847,46.668252]],"type":"Polygon"}

<- Back

Filter Results

Show Full Catalog

☐ Show All imagery in catalog, even imagery I don't have access to order with my current plan

Date range (UTC)

2025/01/01 2025/05/22

High resolution (< 2 m)

☒ SkySat Scene

☒ SkySat Collect

Medium resolution (2-10 m)

☒ PlanetScope Scene

☐ RapidEye Basic Scene

☐ RapidEye Ortho Tile

PlanetScope Only Filters

Spectral bands available

All imagery includes blue, red and green bands

☒ Near infrared

☒ Coastal blue, green II, yellow, red edge

[Learn More](#)

Anomalous Pixels 0.0 - 100.0%

0.0% 100.0%

STAC metadata provides a standardized format for describing geospatial information so that it can be more easily worked with in tool like QGIS.

☐ Add STAC metadata

Review Items

We recommend deselecting items that appear to have no pixels

☒

Draw extent, upload shapefile
Use current displayed extent
Select extent

Filter by date, bands, cloud cover

Select cloud cover, sun azimuth, etc.

Clip item, define what you want to retrieve from the order

2 – Ordering products

Dialog
✕

Order Imagery

1) Name Order
2) Select Assets
3) Tools and Review

Order Name

Order Summary

Order name
firsttry

Number of orders
1

* Each asset will be placed as a separate order

PlanetScope Scene

1 items

Surface reflectance - 4 band

GeoTIFF
 UDM2

Dialog
✕

Order Imagery

1) Name Order
2) Select Assets
3) Tools and Review

1 items

PlanetScope Scene

RECTIFIED ASSETS

Visual
☐

Optimized for visual analysis - RGB only
Bands: blue, green, red

☒ GeoTIFF
☐ NITF

Surface reflectance - 4 band
☒

Corrected for surface reflectance: recommended for most analytic applications - includes RGB NIR
Bands: blue, green, nir, red

☒ GeoTIFF
☐ NITF

UDM2

Surface reflectance - 8 band
☐

Corrected for surface reflectance: recommended for most analytic applications - also includes coastal blue, green II, yellow, red edge
Bands: rededge, green_l, red, blue, coastal, yellow, nir, green

☒ GeoTIFF
☐ NITF

UDM2

Analytic radiance (TOAR) - 4 band
☐

Calibrated to top of atmosphere radiance - includes RGB NIR
Bands: blue, green, nir, red

☒ GeoTIFF
☐ NITF

UDM2

Analytic radiance (TOAR) - 8 band
☐

Calibrated to top of atmosphere radiance - also includes coastal blue, green II, yellow, red edge
Bands: rededge, green_l, red, blue, coastal, yellow, nir, green

Order Summary

Order name
firsttry

Number of orders
1

* Each asset will be placed as a separate order

PlanetScope Scene

1 items

Surface reflectance - 4 band

GeoTIFF
 UDM2

6. Questions

Why does the order have the status "partial"? Does that mean that it is still processing?

When you see "state": "partial" in a Planet Orders API response, it means that the order was only partially successful. Specifically, some of the requested items were delivered, but others were not. This can happen if certain assets were unavailable, you lacked permission to access some items, or for other reasons. The "partial" state is only possible if you set the order type to "partial", which allows the API to deliver whatever items are available and omit those that are not, rather than failing the entire order. You can find more details about which items failed and why in the error_hints field of the response.

If I want to send the same request again but in batches, I won't be charged additionally if it's exactly the same area, right?

With an area under management plan, quota is accounted for as the unionized and dissolved footprints of all the individual requested geometries. If you order the exact same areas, quota will be accounted for these areas only once.

You can re-create the order and introduce a rate limit or split the requests and have the data delivered.

I created some batches now to download the data, and I can see them in My Orders. When I try to download them via my orders download tip I get a message: "there was an error downloading this order". When I click on the specific order name and then download the imagery results it works, but I only see one image in my downloaded folder.

To bundle all results from a Planet Orders API request into a single archive (i.e., a single zip file containing all assets from the order), you need to add the parameter "single_archive": true to your "delivery" object. Your updated delivery payload should look like this:

```
"delivery": {  
  "archive_type": "zip",  
  "single_archive": true,  
  "archive_filename": "batch_37"  
}
```

I added the harmonization tool to the order. When I compare the harmonized one with the PlanetScope data that I downloaded before without harmonization it doesn't seem to be a difference. I evaluated some scatterplots and spectral signatures. Is that expected?

The PlanetScope (PS) constellation includes satellites with different relative spectral responses (RSRs). Using PS data therefore requires an additional harmonization step to compare between different generations of satellites. The harmonize tool allows you to radiometrically harmonize imagery captured by one satellite instrument type to imagery captured by another.

Additionally, the PSB.SD (SuperDove) instrument consists of the next-generation "PSBlue" telescope with a larger 47 megapixel sensor and the same filter response as PS2.SD, in the Red, Green, Blue and NIR bands. The PSB.SD payload extends to include four additional bands; Red Edge, Green I, Yellow and Coastal Blue. Six out of eight bands are meant to be interoperable with Sentinel-2 bands. Therefore, if you are using data from the current SuperDoves, you may notice some changes, but they are very frequently so small that they're not noticeable.

Besides differences in PS RSR, PS surface reflectance (SR) data is often variable for a variety of other reasons. The following items all contribute to PS SR variability:

- directional reflectance effects are not corrected in PS SR data
- stray light can be present in some satellites
- time of day and sun angle vary between scenes
- the radiative transfer models used for SR corrections are based on approximations that may have errors at low sun angles and other adverse conditions

In addition, SR corrections for PS data rely on MODIS estimates of atmospheric optical thickness collected at a different time of day and at a coarse spatial resolution. This leads to a significant degree of variability in PS SR data even within data with the same RSR.

The "harmonize" operation isn't doing only harmonization. It's normalization. In other words, it's a purely empirical correction that's attempting to reduce variability overall.

The impact of the harmonization tool can be more noticeable when dealing with data from different sensors or dates, or when the data has significant scene-to-scene variability.

We downloaded some data for 2025 but realized some weird offset (see attachment).

The image id 20250702_104054_73_24df has ground control as false and is in the preview publishing stage. This image will have an offset due to not being orthorectified per our specification. Approximately 95 percent of groundlock failures are due to cloud cover. However, other factors such as extreme latitudes, challenging topography, and

open water can also affect our ability to georeference images accurately. Non-groundlocked imagery provides only approximate geolocation obtained from the satellite telemetry data. To avoid user confusion or analysis errors, we recommend using imagery that has achieved rectification. Images without ground control points are identified in the metadata with the classification "publishing_stage": "preview".

Please use the image metadata to filter images with ground control true and publishing stage filter set to standard or finalized for you to have correctly orthorectified images.

I would like to know how exactly the compositing is done of the PPlanetScope data. The only information I saw was that within a request the last image is overlaid, no further calculations are made if several images exist?

The composite tool applies images to the composite "in order" with the later images applied over the earlier images. The ordering of the products influences the output where the last item included in the array is applied last on top. No further calculations are applied.

The composite tool is only useful if there are several "incomplete" images for one strip and then they are put together? Is there not just one daily product available that is a composite of all the available scenes on that day?

The composite tool is mainly useful when:

- Your AOI is covered by multiple, partially overlapping scenes (e.g., several "incomplete" scenes per strip or across strips) and you want a single GeoTIFF instead of many tiles.
- You want to optimize quota and file size by merging overlapping scenes into one product.

At this time, there is no single, pre-stitched global image produced for each day. However, we do offer basemaps, which are analysis-ready data products created by mosaicking the best available PlanetScope imagery (weekly, monthly, and quarterly).

I recently discovered that although both images have ground control points and passed all your processing steps, they don't spatially align. There is a small shift. How can that be and, in that regard, also, if a Planetscope image didn't pass the test and landed in quality status "test", how do I know, why that is? Because when I check the criteria, for

example, with the ground control points or the sun elevation and off nadir angle, I don't see why the image didn't pass to "standard".

Small spatial shifts can still occur even when images include Ground Control Points and pass all processing steps. This is expected in some cases and can result from differences in viewing geometry, acquisition conditions, etc. Passing processing means the image meets Planet's accuracy thresholds, but it does not guarantee perfect pixel-level alignment between acquisitions.

Regarding images with Quality Status "test": this status is based on a combination of internal automated quality checks. An image may meet visible metadata criteria (such as GCPs, sun elevation, or off-nadir angle) but still fail other internal checks or a combined confidence assessment, leading it to be classified as "test" rather than "standard."

